

Approach To Puberty , Disorders Of Sex Development And Childhood Gynecological Disorders

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Objectives

- Describe the normal changes of puberty and the secondary sexual differentiation that accompanies it.
- Understand the classification and causes of abnormal puberty and disorders of sexual development (DSD).
- Understand the common gynecological problems happening during childhood

Normal puberty

- Puberty is the process of reproductive and sexual development and maturation that changes a child into an adult.
- During childhood, the HPO axis is suppressed and levels of GnRH, FSH and LH are very low.
- From the age of 8–9 years GnRH is secreted in pulsations of increasing amplitude and frequency, these are initially sleep-related, but as puberty progresses, these extend throughout the day.
- .

- This stimulates secretion of FSH and LH by the pituitary glands, which in turn triggers follicular growth and steroidogenesis in the ovary.
- The oestrogen produced by the ovary then initiates the physical changes of puberty.
- The exact mechanism determining the onset of puberty is still unknown, but it is influenced by many factors including race, heredity, body weight and exercise.
- Kisspeptin, leptin and many other puberty activating /inhibitor genes are involved in the onset of puberty

- The physical changes occurring in puberty are
 - 1-breast development (thelarche).
 - 2- pubic and axillary hair growth (adrenarche).
 - 3- growth spurt.
 - 4-onset of menstruation (menarche).
- The first physical signs of puberty are breast budding and this occurs 2–3 years before menarche.
- The appearance of pubic hair is dependent on the secretion of adrenal androgens and is usually after thelarche.

- growth hormone secretion also increases, leading to a pubertal growth spurt.
- The mean age of menarche is 12.8 years and it may take over 3 years before the menstrual cycle establishes a regular pattern.
- Pubertal development was described by Tanner and the stages of breast and pubic hair development are often referred to as Tanner stages 1–5

Stage	Pubic hair scale	Breast development
Stage 1	No hair	No glandular breast tissue palpable
Stage 2	Downy hair	Breast tissue palpable under the areola (the 1st pubertal sign in female)
Stage 3	Scant terminal hair	Breast tissue palpable out side the areola; no areolar development
Stage 4	Terminal hair that fills the entire triangle overlying the pubic region	Areola elevation above the contour of the breast, forming a double a double scope appearance.
Stage 5	terminal hair that extends beyond the inguinal crease onto the thigh	Areolar mound recedes into single breast contour with areolar hyper pigmentation ,papillae development and nipple protrusion.

Tanner staging.

	1	2	3	4	5
Breast	 Prepubertal	 Breast and papilla are elevated as a small mound. Areolar diameter increases	 Further enlargement of the breast bud with loss of the contour separation between breast and areola	 Aerolar and papilla form a secondary mound	 Mature areolar is part of the general breast contour
Pubic hair	 Prepubertal	 Sparse lightly pigmented chiefly along the medial border of the labia majora	 Darker beginning to curl, increased amount spreading over the mons	 Increased amount of course, curly but limited to the mons	 Adult feminine triangle with spread to the medial surface of the thighs

Precocious puberty

- This is defined as the onset of puberty before the age of 8 in a girl or 9 in a boy.
- Central precocious puberty is gonadotrophin dependent. The aetiology is often unknown, although up to 25% are due to central nervous system (CNS) malformations or brain tumours.
- Peripheral precocious puberty, which is gonadotrophin independent, is always pathological and can be caused by oestrogen secretion, such as exogenous ingestion or a hormone-producing tumour.

Delayed puberty :

there are no signs of secondary sexual characteristics by the age **14** years.

Its either due to :

1- hypogonadotrophic- hypogonadism:

- this may be central or constitutional, but other causes may be excluded. These include anorexia nervosa, excessive exercise and chronic illness, such as diabetes or renal.
- Rarer causes include a pituitary tumour and kallmann syndrome.
- it is associated with delayed puberty and primary amenorrhea.

2-hypergonadotrophic hypogonadism:

- this is caused by gonadal failure.
- the gonad does not function, despite high gonadotropins .
- it is associated with Turner syndrome and XX gonadal dysgenesis.
- premature ovarian failure can occur at any age, including prior to pubertal age, and may be idiopathic, but can also be part of an autoimmune or metabolic disorder or following chemo- or radiotherapy for childhood cancer.

Disorders of sexual development DSD:

- DSD are conditions occur when there is disruption of either: *Gonadal differentiation and Fetal sex steroid production or action.*
- The clinical consequences of this depend upon where within the sequence the variation occurs.
- DSD may be diagnosed at birth with ambiguous or abnormal genitalia, but may also be seen at puberty in girls who present with primary amenorrhoea or increasing virilization.

Table 3.2 Summary of terminology for disorders of sex development (DSDs) according to the 2006 Chicago consensus statement

Previous terms	Accepted revised terms
Intersex	DSD
Male pseudohermaphrodite	46,XY DSD
Undervirilization of XY male	
Undermasculinization of XY male	
Female pseudohermaphrodite	46,XX DSD
Overvirilization of an XX female	
Masculinization of an XX female	
True hermaphrodite	Ovotesticular DSD
XX male or XX sex reversal	46,XX testicular DSD
XY sex reversal	46,XY complete gonadal dysgenesis

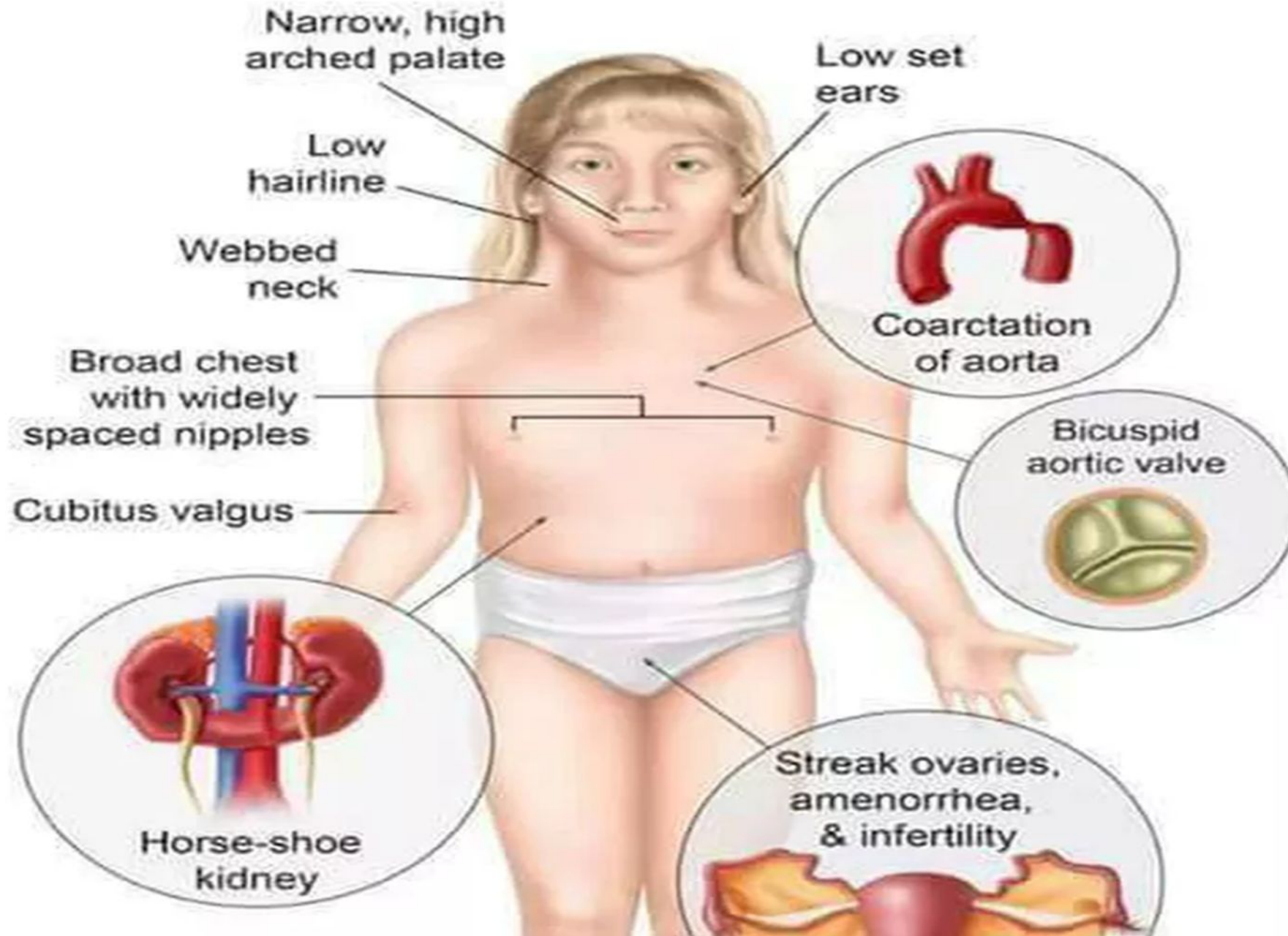
Source: Hughes *et al.* (2006).

Non-structural causes of DSD (chromosomal)

Turner syndrome

- The total complement of chromosomes is 45 in Turner syndrome, which results from a complete or partial absence of one X chromosome (45XO).
- Turner syndrome is the most common chromosomal anomaly in females, occurring in 1 in 2,500 live female births.
- A mosaic karyotype is not uncommon, leading to a variable presentation.

Turner syndrome

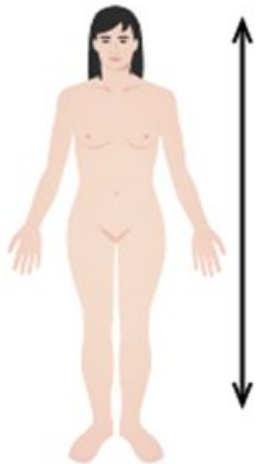


- In this condition, the ovary does not complete its normal development and only the stroma is present at birth (streak gonads)
- In childhood, treatment is focused on growth, but in adolescence it focuses on induction of puberty.
- Pregnancy is only possible with ovum donation. Psychological input and support is important.
- In girls with mosaicism the clinical picture can vary and normal puberty and menstruation can occur, with early cessation of periods.

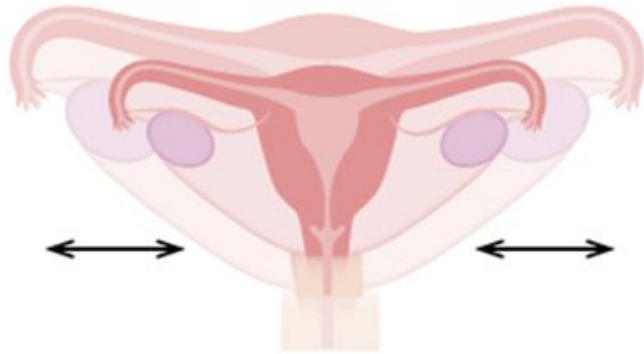
46XY gonadal dysgenesis:

- In this situation, the gonads do not develop into a testis, despite the presence of an XY karyotype., In about 15% of cases, this is due to a mutation in the SRY gene on the Y chromosome.
- In complete gonadal dysgenesis (Swyer syndrome), the gonad remains as a streak gonad and does not produce any hormones.
- In the absence of (AMH), the Müllerian structures do not regress and the uterus, vagina and Fallopian tubes develop normally.
- The absence of testosterone means the fetus does not virilize.
- The baby is phenotypically female, although has an XY chromosome.

- The gonads do not function and presentation is usually at adolescence with delayed puberty.
- The dysgenetic gonad has a high malignancy risk and should be removed when the diagnosis is made.
- Puberty must be induced with oestrogen and pregnancies have been reported with a donor oocyte.
- Full disclosure of the diagnosis including the XY karyotype is essential, although this can be devastating and specialized psychological input is crucial.



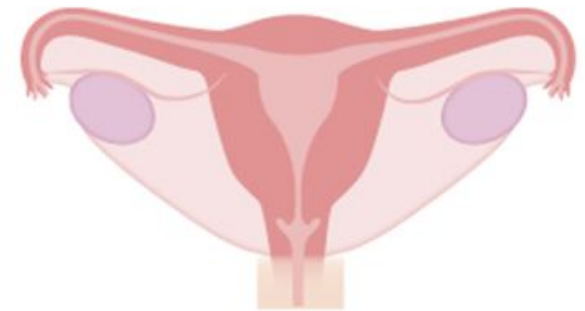
Tall
stature



Small
uterus



No sexual
characteristics

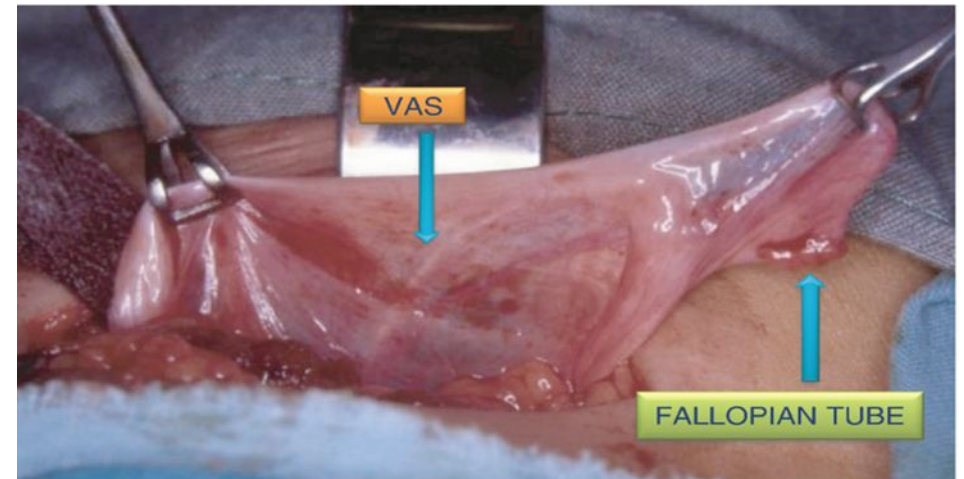
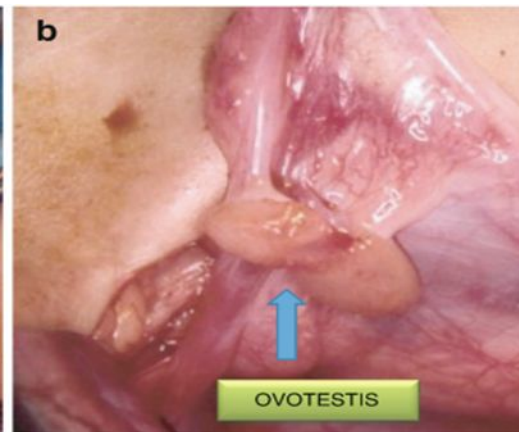
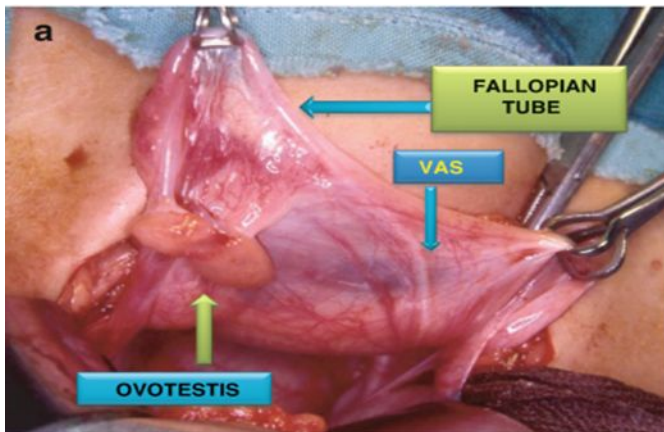


No
ovaries

Ovo testicular DSD

Mixed gonadal dysgenesis is a more complex condition. The karyotype may be 46XX, but XX/XY mosaicism is present in up to 20%. In this situation, both functioning ovarian and testicular tissue can be present.

The anatomical findings vary depending on the function of the gonads.



46XY DSD

- The most common cause of 46XY DSD, complete androgen insensitivity syndrome (CAIS), occurs in individuals where virilization of the external genitalia does not occur, due to a partial or complete inability of the androgen receptor to respond to androgen stimulation.
- In the fetus with CAIS, testes form normally due to the action of the SRY gene.
- Hence, CAIS women do not have a uterus (AMH is present)
- Testosterone is also produced at the appropriate time; however, due to the inability of the androgen receptor to respond, the external genitalia do not virilize and instead undergo female development.
- The baby is born with normal female external genitalia.

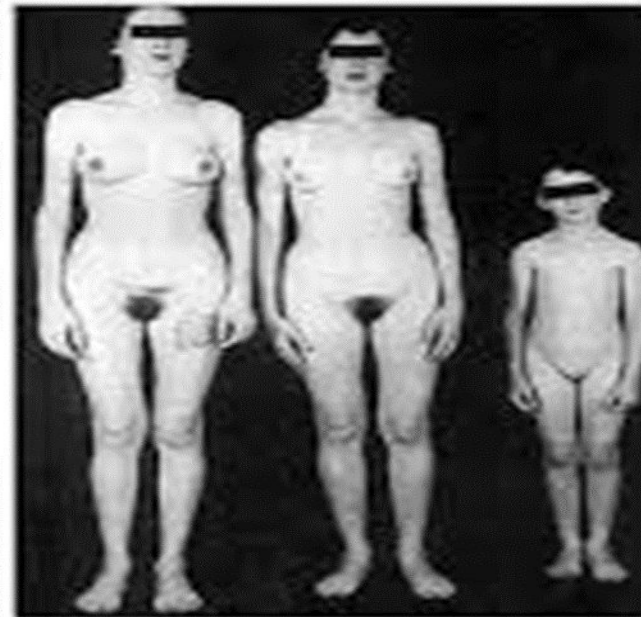
- Presentation is usually at puberty with primary amenorrhoea, although if the testes are in the inguinal canal they can cause a hernia in a younger girl.
- Once the diagnosis is made, initial management is psychological with full disclosure of the XY karyotype and the information that the patient will be infertile.
- Gonadectomy is recommended because of the small long-term risk of testicular malignancy, although this can be deferred until after puberty.
- Once the gonads are removed, long-term HRT will be required.

In cases of **partial androgen insensitivity**, the androgen receptor can respond to some extent with limited virilization. The child is usually diagnosed at birth with ambiguous genitalia.



Testicular feminization—Bilateral inguinal gonads

Androgen-Insensitivity Syndrome



- Genetically male (XY)
- Testosterone is secreted
- Target cells lack receptors for the hormone
- No masculinizing effects occur

5-Alpha-reductase deficiency:

- In this condition, the fetus has an XY karyotype and normal functioning testes that produce both testosterone and AMH.
- The fetus is unable to convert testosterone to dihydrotestosterone in the peripheral tissues and so cannot virilize normally.
- Presentation is usually with ambiguous genitalia at birth, but can also be with increasing virilization at puberty of a female child.

46XX DSD:

- The most common cause of 46XX DSD, **congenital adrenal hyperplasia (CAH)**, leads to virilization of a female fetus.
- It is due to an enzyme deficiency in the corticosteroid production pathway in the adrenal gland, with over 90% being a deficiency in 21-hydroxylase, which converts progesterone to deoxycorticosterone and 17-hydroxyprogesterone (17-OHP) to deoxycortisol.
- The reduced levels of cortisol being produced drive the negative-feedback loop, resulting in hyperplasia of the adrenal glands, This leads to an excess of androgen precursors and then to elevated testosterone production.

- 2/3 of children with 21-hydroxylase CAH will have a 'salt-losing' variety, which also affects the ability to produce aldosterone which represents a life threatening situation, and those children who are salt-losing often become dangerously unwell within a few days of birth.
- Affected individuals require life-long steroid replacement, such as hydrocortisone, along with fludrocortisone for salt losers.
- Once the infant is well and stabilized on their steroid regime, surgical treatment of the genitalia is considered.

- This management is now controversial as adult patients with CAH are very dissatisfied with the outcome of their surgery and argue that surgery should have been deferred until they were old enough to have a choice.



FIG. 594.2 Three virilized females with untreated congenital adrenal hyperplasia. All were erroneously assigned male sex at birth, and each had a normal female sex-chromosome complement. Infants A and B had the salt-wasting form and received the diagnosis early in infancy. Infant C was referred at 1 yr of age because of bilateral cryptorchidism. Notice the

Gynaecological disorders in children :

- The disorders fall into two groups: those related to pre-puberty and those of adolescence.
- primarily involve vulvovaginitis, although the presence of foreign bodies in the vagina and vaginal bleeding and labial adhesions should also be considered.
- The adolescent is usually troubled primarily by menstrual disorders, including heavy uterine bleeding, irregular menstruation, primary dysmenorrhoea and hirsutism.

Vulvovaginitis

- Its aetiology is based on opportunistic bacteria colonizing the lower vagina and inducing an inflammatory response.
- At birth the vulva and vagina are well oestrogenized due to the intrauterine exposure of the fetus to placental oestrogen. However, within 2–3 weeks of delivery the resultant hypo-oestrogenic state leads to changes in the vulval skin and the vaginal epithelium become much thinner.(no labtobacilli and low PH)
- The lack of labial protection and the close apposition of the anus mean that the vulva and lower vagina are constantly exposed to faecal bacterial contamination.

- Vulvovaginitis may also occur in childhood in those who have an impaired local host defenses deficiency.

Causes of vulvovaginitis in children.

- Bacterial: Non-specific (common) and Specific (rare)
- Fungal (rare): Candida of vulva only
- Viral (rare) may be due to sexual abuse.
- Dermatitis : Atopic, Lichen sclerosis and Contact
- Sexual abuse
- Enuresis
- Foreign body

- Lichen sclerosis may cause persistent vulval itching, the skin undergoes atrophy and fissuring and is very susceptible to secondary infection.
- Enuresis creates a moist vulva allowing secondary infection by bacteria leading to vulvovaginitis.
- The diagnosis will be by inspection of the vulva and vagina with good illumination also if there is possibility of foreign body in the vagina the examination will be by using otoscope through the hymen. Also bacteriological sample from the discharge.
- In some time pinworms should to be excluded .

Treatment of vulvovaginitis:

- After micturition the mother and child should be instructed to clean the vulva completely and not to leave the vulval skin wet, as this damp warm environment is an ideal culture surface for bacteria that cause vulvovaginitis.
- Vulval hygiene through daily washing by gentle soap.
- Excessive washing of the vulva must be avoided.
- In the case of specific organisms being identified antibiotics can be prescribed and amoxicillin is probably the most effective.
- Barrier cream.

Foreign body

- Foreign bodies are occasionally found in the vagina and may lead to vaginal discharge.
- Ultrasound scan may detect a foreign body .
- If a history of a foreign body is forthcoming, it is probably best to carry out an examination under anaesthetic and remove any foreign body at that time.

Vaginal bleeding

- extremely rare and should always be treated with extreme caution.
- The causes of genital bleeding in childhood include a vaginal foreign body, trauma, a neoplasm, premature menarche or urethral prolapse and the diagnosis can almost always be made on clinical inspection.

Labial adhesions:

- Usually an innocent finding and a trivial problem, but its importance is that it is frequently misdiagnosed as congenital absence of the vagina.
- It happens due to hypo-oestrogenic state; the labia minora stick together in the midline, usually from posterior to anterior until only a small opening is left through which urine is passed.
- In the majority of cases no treatment is required.
- local oestrogen cream can be applied for about 2 weeks after that usually complete resolution of the labial adhesions.
- In some rare cases gentle separation of the labia may be undertaken using a probe.

- Application of a bland barrier cream at this stage will prevent further adhesion formation.
- Very rarely labial adhesions may be the result of sexual abuse.



Fig. 1 Clinical picture of typical pre-pubertal synechia of the labia in a 5-year-old girl.

Menstrual disorders in adolescents are usually a reflection of normal physiology.

- Treatment should only be instigated if the adolescent is found to be anaemic.
- Treatments should be as simple as possible.

Hirsutism in adolescents is often idiopathic or cultural.

Treatment, as in the adult, is by cosmetic means or endocrine manipulation.

Causes of hirsutism in adolescents.

- Congenital adrenal hyperplasia (Classic and Late onset
- Androgen-secreting tumours
- Polycystic ovarian syndrome
- Idiopathic
- XY gonadal dysgenes

Thank you

Refences :is textbook gynecology by ten teacher 21st edition











